

Daily Geometry Drill #1

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Gradient, intercept, midpoint, distance — read them straight off. No working shown needed.

- Find the gradient of the line $6x - 3y + 15 = 0$.
- Find the gradient of the line through $(1, 2)$ and $(5, 14)$.
- A line of gradient 4 is perpendicular to a line of gradient m . Write down m .
- Find the distance between $(2, 1)$ and $(5, 5)$.
- Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.
- Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.
- A line has x -intercept 4 and y -intercept 6. Find its gradient.
- Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.
- The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .
- Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Section B Manipulation Drills

(≤50 s each)

Build the line, then interrogate it. Extra tool required per question.

- The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

A) $y = -\frac{3}{2}x + 1$

B) $y = \frac{3}{2}x - 2$

C) $y = \frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + 3$

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

A) 1

B) 2

C) 3

D) 4

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

A) $(2, 5)$

B) $(3, 3)$

C) $(4, 4)$

D) $(5, 3)$

4. Find the area of the quadrilateral with vertices $(0, 0)$, $(4, 0)$, $(5, 3)$, $(1, 3)$, in that order.

A) 10

B) 12

C) 14

D) 16

5. The line through $(1, 2)$ and $(4, 5)$ crosses the y -axis at which value?

A) 0

B) 1

C) 2

D) 3

6. Find the perpendicular distance from $(0, 1)$ to the line $6x - 8y + 2 = 0$.

7. Which of the following is NOT parallel to $y = 3x + 1$?

A) $y = 3x + 5$

B) $6x - 2y = 4$

C) $3x - y = 2$

D) $y = 2x + 1$

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .

9. Which point is closest to the origin?

- A) (1, 3)
- B) (2, 2)
- C) (4, 0)
- D) (0, 5)

10. Using the shoelace formula, find the area of the triangle with vertices (1, 1), (4, 2), (2, 6).

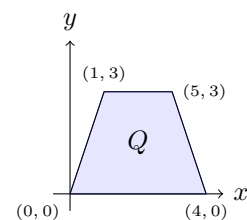
Section C Substitution & Structure

(≤90 s each)

Spot the tool first — the numbers were chosen so that a clean route exists. These are genuinely harder: non-obvious structure, hidden invariants, and competition traps.

1. The lines $l_1 : y = 6 - 2x$ and $l_2 : y = \frac{1}{2}x - 4$ are perpendicular and intersect at P . The three lines l_1 , l_2 and the x -axis enclose a triangle T . What is the area of T ? (after TMUA 2017 Paper 1 Q3)
 - A) $\frac{81}{5}$
 - B) $\frac{54}{5}$
 - C) $\frac{81}{10}$
 - D) $\frac{27}{2}$
2. The perpendicular bisector of the segment joining $A(2, -6)$ and $B(5, 4)$ meets the x -axis at P . What is the x -coordinate of P ? (after TMUA Specimen Paper 1 Q3)
 - A) $\frac{1}{6}$
 - B) $\frac{1}{3}$
 - C) $\frac{19}{5}$
 - D) $\frac{41}{6}$
3. Two circles have the same radius r . Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which line is the common chord PQ ? (adapted from TMUA 2021 Paper 1 Q1)
 - A) $5x - 3y = 1$
 - B) $5x - 3y = 4$
 - C) $5x + 3y = 1$
 - D) $3x - 5y = 4$
4. For which values of p does $x^2 - 2px + y^2 - 6y - p^2 + 8p + 9 = 0$ represent a (real) circle? (after TMUA 2022 Paper 1 Q2)
 - A) $p < -1$ or $p > 9$
 - B) $-1 < p < 9$
 - C) $0 < p < 4$
 - D) $p < 0$ or $p > 4$

5. The points $A(1, 4)$, $B(4, 7)$ and $C(10, k)$ are collinear. The line through C perpendicular to AB meets the y -axis at D . What is the y -coordinate of D ?
- A) 23
B) 24
C) 25
D) 26
6. The curve $y = 4x^2$ is translated by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$, then reflected in the x -axis, then stretched parallel to the x -axis with scale factor 2. What is the equation of the final curve? (*adapted from TMUA 2020 Paper 1 Q10*)
- A) $y = -x^2 + 12x - 31$
B) $y = -x^2 + 12x - 41$
C) $y = -16x^2 + 48x - 31$
D) $y = 16x^2 - 48x + 41$
7. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (*adapted from TMUA 2019 Paper 1 Q6*)
- A) 8
B) 16
C) 26
D) 42
8. The quadrilateral Q has vertices $(0, 0)$, $(4, 0)$, $(5, 3)$ and $(1, 3)$ in that order (a parallelogram). A second quadrilateral Q' is obtained by reflecting Q in the line $y = x$. What is the area of the overlap $Q \cap Q'$?
- A) 6
B) 8
C) 9
D) 12

**Section D Challenge**

(TMUA / SMC / BMO1 difficulty)

Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.

1. In $\triangle ABC$, $AB = 10$, $BC = 7$ and $\angle A = \theta$. There are two distinct non-congruent triangles T_1 (larger area) and T_2 (smaller area) satisfying these data. It is known that $\text{Area}(T_1) = 3 \text{Area}(T_2)$. Find $\cos \theta$. (*adapted from TMUA 2018 Paper 1 Q19 / BMO1 2015 Q3*)
- A) $\frac{5}{7}$
B) $\frac{\sqrt{17}}{5}$

C) $\frac{\sqrt{51}}{8}$

D) $\frac{\sqrt{34}}{8}$

2. The square $MNOP$ has perimeter 40 (so side 10). Points R, S, T, U lie on MN, NO, OP, PM respectively with $MR = x$ and $RSTU$ a rectangle whose sides are at 45° to the square's sides. For $0 < x < 10$, the area of $RSTU$ is $A(x) = 20x - 2x^2 + 100 - 20x$ (simplified to $A = -2x^2 + 20x$ up to translation). What is the *largest* x for which $A(x) = 20$? (after TMUA 2023 Paper 1 Q5)

A) $5 + \sqrt{5}$

B) $5 + \sqrt{15}$

C) $5 + \sqrt{20}$

D) $10 - \sqrt{5}$

3. A circle has centre O and radius 6. Points P, Q, R lie on the circle with $\angle POQ \geq \frac{\pi}{2}$ and $[POQ] = 9\sqrt{3}$ (area). What is the *greatest possible* area of $\triangle PRQ$? (adapted from TMUA 2022 Paper 1 Q14)

A) $18 + 9\sqrt{3}$

B) $27\sqrt{3}$

C) $27 + 9\sqrt{3}$

D) $36 + 9\sqrt{3}$

4. Let $P(p, q)$ and circle $C : x^2 + 2fx + y^2 + 2gy + h = 0$ with centre $C_0 = (-f, -g)$ and radius $r = \sqrt{f^2 + g^2 - h}$. Let $L = |PC_0|$. Which data set is *minimal sufficient* to determine L ? (adapted from MAT 2020 Q1H / TMUA 2022 Paper 2 Q4)

A) f, g, h

B) f, g, p, q

C) f, h, p, q

D) g, h, p, q

5. Circle $C_1 : x^2 + y^2 = 25$ is fixed. A second circle C_2 has radius 4 and centre (a, b) uniformly random in the rectangle $-2 \leq a \leq 2, -3 \leq b \leq 3$ (so the centre is equally likely to be anywhere in that 4×6 rectangle). What is the probability that C_1 and C_2 intersect (in two points, touching counts as intersecting)? (after TMUA 2022 Paper 1 Q19)

A) $\frac{9}{25}$

B) $\frac{16}{25}$

C) $\frac{16-\pi}{24}$

D) $\frac{24-\pi}{24}$

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths